

A/B Testing & Hypothesis Testing — Checkout Conversion

Data note: No source experiment dataset was supplied. This report uses an illustrative 10,000-session dataset (5,000 Control and 5,000 Variant) to demonstrate the complete statistical workflow.

1. Experiment Design

Control (A): existing checkout experience. Variant (B): proposed checkout experience. Primary metric: checkout conversion rate. Significance level $\alpha = 0.05$.

2. Hypotheses

H0: $p_B = p_A$ (no difference in conversion rate).

H1: $p_B \neq p_A$ (conversion rates differ).

Metric	Control A	Variant B
Sample size	5,000	5,000
Conversions	450	550
Conversion rate	9.00%	11.00%

3. Statistical Test

Two-sample proportion Z-test: $z = 3.3333$; two-sided p-value = 0.000858121. A chi-square test of independence gives $\chi^2 = 11.1111$, $p = 0.000858121$. The two tests agree for this 2x2 conversion table.

4. Effect Size & Confidence Interval

Absolute conversion lift = 2.00% (Variant – Control). Relative lift = 22.22%. Cohen's $h = 0.0667$. 95% confidence interval for the absolute lift = [0.82%, 3.18%].

5. Sample Size

Using the 9% baseline and targeting a 20% relative lift (10.8% Variant), an approximate 80%-power calculation requires about 4,321 observations per arm under the stated assumptions. The illustrative experiment has 5,000 per arm, so this calculation should be treated as planning guidance rather than a post-hoc validity claim.

6. Executive Decision Memo

Statistically, the observed conversion-rate difference is significant at $\alpha = 0.05$ because $p = 0.000858121$ and the 95% CI for the lift excludes zero. The data show a 22.2% relative increase in conversion in the illustrative sample. A production decision should additionally consider experiment integrity, guardrail metrics, persistence of the effect, and financial impact before rollout.

7. Limitations

This is an illustrative dataset because no source A/B test data were provided. The result is not evidence about a real website. In a real experiment, verify random assignment, sample-ratio mismatch, exposure counts, experiment duration, independence assumptions, and pre-specified primary/guardrail metrics.